



Developing a Road Exposure Index for Herder–Farmer Conflicts in Nigeria’s Nomadic Pastoral Corridors

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Abstract: This study evaluates the spatial exposure of road infrastructure and road users to nomadic herder–farmer conflicts (HFCs) in Nigeria, focusing on four critical nomadic pastoral corridors (NPCs): Kaduna–Plateau (North West–North Central, NW–NC), Taraba–Benue (North East–North Central, NE–NC), Nasarawa–Benue (North Central–North Central, NC–NC), and Benue–Enugu (North Central–South East, NC–SE). Using geospatial data, the study develops a road exposure index (REI) by integrating road density, population density, and conflict density across local government areas within these NPCs. Kernel density estimation (KDE) was applied in ArcGIS to visualize the spatial distribution of REI values and identify areas of heightened exposure during HFC-affected years. The results reveal substantial variations in REI across the corridors, with tertiary roads in the Benue–Enugu (NC–SE) and Nasarawa–Benue (NC–NC) corridors showing particularly high levels of exposure. These areas are important because they include major food-producing zones, suggesting that HFC-related road exposure may have implications for food accessibility and rural–urban market connectivity. The findings highlight the need for targeted intelligent transport system (ITS) infrastructure and security interventions to improve road monitoring, enhance transport resilience, and reduce mobility risks in conflict-affected corridors in Nigeria.

Keywords: Herder–farmer clashes; Conflict corridor model; Road accessibility; Food security; Intelligent transport systems

1 Introduction

Globally, transport networks and users are increasingly concentrated in environments where interacting pressures amplify their exposure to hazards [1, 2]. Road networks, which are embedded within complex socio-environmental systems, are therefore susceptible to both direct attacks and indirect stressors associated with natural hazards, armed conflict, civil unrest, and other forms of instability that disrupt mobility and reduce accessibility [3–9]. In particular, disruptions caused by conflict and insecurity can undermine the use of critical roads, complicate emergency response, disrupt development continuity, and reshape accessibility and movement patterns [1, 4, 5, 9].

This issue is especially critical in Africa, where road infrastructure is often inadequate, poorly maintained, and insufficiently connected to settlements. In many areas, roads are also used by both inter-state vehicular traffic and nomadic pastoral movement. In this context, herder–farmer conflicts (HFCs), which arise from interactions between herding and farming communities, can significantly affect paved road networks and users, especially roads connecting different settlements. According to Faiyetole and Abiodun [10], these conflicts may be driven by climate change, land-use change, and sociopolitical triggers, including the politicization of ethnic identity [11].

Pastoral mobility and farming activities once coexisted relatively peacefully. However, as Faiyetole and Abiodun [10] noted, additional pressures from environmental degradation, population growth, and weaknesses in governance and institutions have transformed these formerly harmonious interactions into widespread HFCs that now affect every geopolitical zone in Nigeria [12–14]. As competition for natural resources intensifies, the boundaries between farming and grazing lands become increasingly unclear, leading to persistent friction and recurrent violence [15–17]. These conflicts increasingly extend beyond traditional rural settings into the federal

road system, peri-urban fringes, and public and private infrastructure. As suggested by Jenelius and Mattsson [8] and Nwankwo [18], the destructive reach of HFCs highlights the importance of assessing road network exposure in conflict-prone environments, a concern further supported by Alkama et al. [19].

Previous studies have largely examined road vulnerability through network performance metrics such as accessibility loss, detour costs, and connectivity disruption [2, 8]. While valuable, such approaches are less suited to conflict-affected rural contexts owing to the need for detailed traffic data which are unavailable and where the primary concern is not network efficiency but road users' exposure to HFCs. Therefore, studies such as Brottem [15] and Koks et al. [9] emphasize the need for spatially explicit approaches that capture the concentration of conflict pressures and infrastructural exposure rather than system-wide performance loss.

Broadly, studies assessing road-related risk generally fall into two methodological strands. The first comprises network vulnerability and performance-based models, which simulate link failure and quantify impacts on accessibility, detour costs, or travel-time loss [2, 8, 9, 20]. These approaches are analytically rigorous but require detailed traffic flow and origin–destination (OD) data that are often unavailable in rural, HFC-infested areas. The second strand consists of spatial exposure and composite indicator approaches, which overlay hazard intensity with infrastructure and population layers to identify hotspots [3, 21–23]. While suitable for screening purposes, most applications focus on natural hazards and are not tailored to corridor-scale conflict dynamics. Within this context, the contribution of the present study is the development of an HFC-specific, corridor-scale exposure index using open geospatial data, enabling the identification and ranking of roads most exposed to HFC pressures. To support the novelty claim, Table 1 positions road exposure against dominant road vulnerability and risk-modelling families and highlights differences in outcomes and data requirements.

Table 1. Key parameters of our model

Methodological Strand	Focus	Data Requirements	Output	Limitations	Representative Studies
Network vulnerability / performance models	Accessibility loss, detour cost, road conditions or road system performance under link disruption	Traffic flows, origin–destination (OD) matrices, calibrated networks	Travel-time delay, detour cost such as distance and fuel costs, critical link ranking	Data-intensive; difficult in rural, conflict-affected regions	[2, 4, 5, 8, 9, 20, 24–30]
Spatial exposure / composite indicator approaches	Spatial overlap of hazard, infrastructure, and population	Hazard maps, infrastructure and demographic layers	Hazard exposure hotspot maps and index scores	Primarily natural hazard focused; limited use of conflict corridor model (CCM)	[3, 21–23, 31–36]
Road exposure framework	Conflict-specific, corridor-scale exposure screening	Conflict density, road density, population density	Exposure intensity and corridor ranking	Based on the CCM approach; does not model road vulnerability or transport network risk	This study

Building on the frameworks of the United Nations Office for Disaster Risk Reduction (UNDRR) [37] and the Intergovernmental Panel on Climate Change (IPCC) [38], this study develops a conceptual framework for assessing road exposure to HFCs (Figure 1). Within this framework, hazard is operationalized as the occurrence and intensity of HFCs within nomadic pastoral corridors (NPCs), as identified by Faiyetole and Abiodun [10] using the Conflict Corridor Model (CCM) across Nigeria's six geopolitical zones. Exposure to HFC hazards is determined by the presence of road infrastructure and road users. Road infrastructure is represented by road density, while road users are represented by population density within the NPCs.

Accordingly, the road exposure index (REI) is constructed from three components that align with exposure-based assessment: (i) road density as a proxy for the infrastructure footprint within the NPCs, (ii) population density as a proxy for the scale of human presence and potential societal disruption, and (iii) HFC density as a proxy for hazard intensity and recurrence within the NPCs. Road vulnerability and transport network risk are shown in the conceptual framework as related but excluded components because they are beyond the scope of this study. These components would require additional indicators, such as road condition, network redundancy, security response capacity, emergency response capacity, road closure, travel-time delay, fatalities, and economic losses within the

corridor.

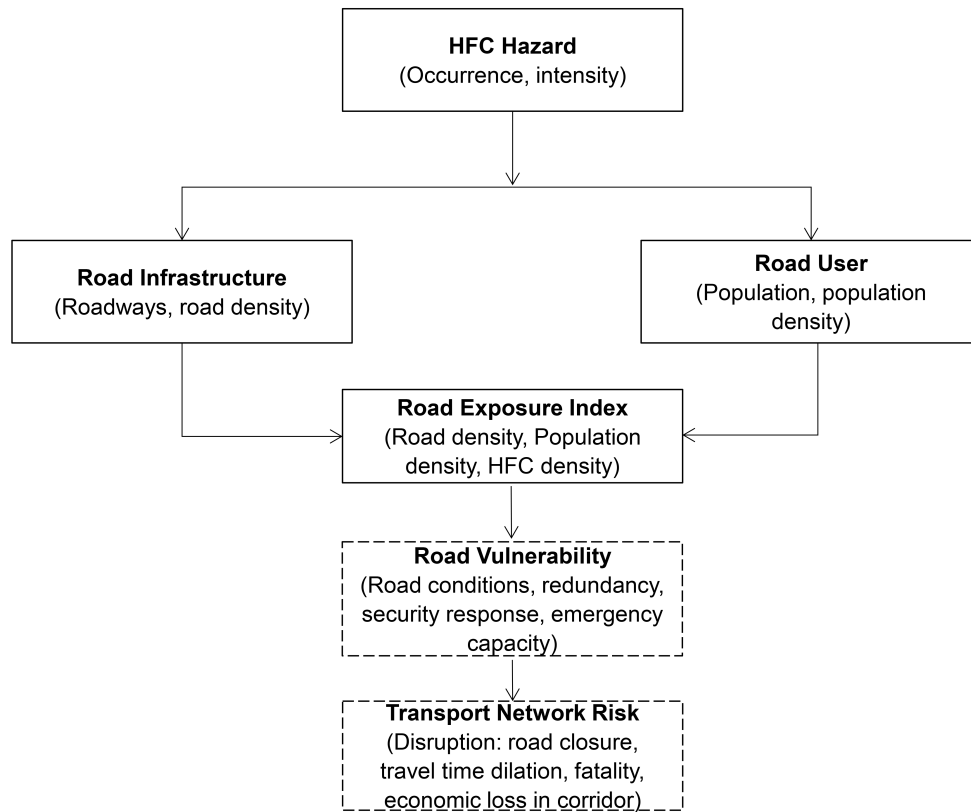


Figure 1. Conceptual framework of herder–farmer conflict (HFC) hazard and the road exposure index (REI)

REI is therefore a socio-spatial exposure index for assessing the exposure of road infrastructure and users to HFC hazards. It addresses a critical methodological gap in assessing transport exposure in HFC-prone environments by focusing on spatially concentrated insecurity rather than classical network vulnerability. Rather than modelling network vulnerability in the classical engineering sense, the REI identifies areas where road infrastructure and users are most exposed to HFCs.

Using population density, road density, and HFC incidence as spatial indicators, the approach captures the intensity and spatial concentration of exposure across key NPCs using CCM, including the Kaduna–Plateau, Benue–Enugu, Nasarawa–Benue, and Taraba–Benue NPCs. Therefore, this study aims to quantify and classify the roads along these NPCs that are most exposed to nomadic HFCs in Nigeria. By distinguishing road exposure from classical road vulnerability, the study provides a screening framework for identifying where conflict hazards, road infrastructure, and road users spatially converge in data-scarce rural corridors.

2 Methodology

2.1 Operationalization of the Road Exposure Index Model

Road exposure is operationalized as the socio-spatial intersection of road infrastructure and road users with conflict hazards following the UNDRR [37] multidimensional risk frameworks used in fragile, high-risk environments [3, 7, 32, 35, 39].

Building on the CCM developed by Faiyetole and Abiodun [10], this study focuses on four key NPCs, which serve as hotspot areas where HFCs intersect with road networks (see Figure 2). For clarity, the geopolitical-zone abbreviations used here are North West (NW), North Central (NC), North East (NE), and South East (SE). The HFC data cover the period from 2015 to 2023 and were sourced from Nigeria Watch [40]. State and ward shapefiles, together with population data, were obtained from GRID3 [41] and City Population [42], while road shapefiles were acquired from OpenStreetMap (OSM) using the QuickOSM plugin in QGIS.

OSM data quality, completeness, and validation: OSM is the only globally available open-access road network dataset with consistent classification attributes, and its overall coverage exceeds 80% worldwide [43]. However, data completeness varies spatially, particularly between urban and rural environments. While positional accuracy of mapped roads is generally high, completeness of minor and tertiary roads is more variable, especially in rural and data-scarce regions [44]. This issue is directly relevant to the present study, as the four pastoral corridors traverse

predominantly rural and peri-urban landscapes. In such settings, tertiary and unclassified roads, which correspond to low-order rural access routes, may be underrepresented due to uneven volunteer mapping intensity [44–47]. Consequently, OSM-derived road density may underestimate the absolute density of minor rural routes in certain wards [46]. However, trunk, primary, and secondary road classes which constitute the principal structural network of corridor connectivity are typically more consistently mapped and exhibit higher positional reliability [43, 46, 48].

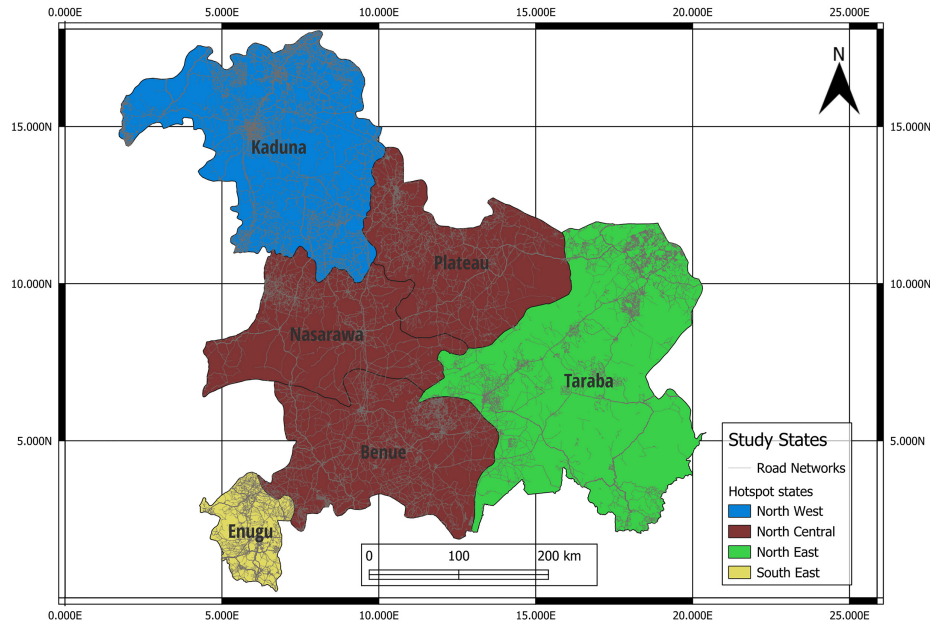


Figure 2. Map of the study areas in Nigeria, reproduced from Ref. [10] under the Creative Commons CC BY 4.0 license

To mitigate data-quality concerns within the constraints of data-scarce environments, validation and consistency checks were undertaken. Extracted road geometries and highway classifications were visually cross-checked against high-resolution satellite imagery in ArcGIS Pro and Google Earth Pro using stratified sampling across major and minor road classes within each corridor. This approach confirmed strong positional alignment for primary and secondary road classes and identified limited but expected gaps in minor rural access routes, consistent with findings that OSM performs well for higher-order roads but less reliably for tertiary networks [46, 49]. Geometry topology checks were subsequently performed to remove duplicates and repair line errors prior to density calculation. Final road layers were clipped to corridor boundaries to prevent edge inflation in density estimation.

Population data harmonisation and temporal alignment: Population data were derived from the GRID3 [41] programme, which provides geo-referenced administrative-level population estimates for Nigeria. GRID3 was used as the baseline spatial population distribution across wards. To account for temporal variation between 2015 and 2023, annual growth rates and official projections from City Population [42] were applied to scale the baseline ward population values. This procedure helps reduce inconsistencies caused by combining independent spatial population datasets and improves temporal comparability across corridors and conflict years.

Consequently, to quantify the REI per NPC, road density, population density and conflict density were used. The selection of road density, population density, and conflict density as core indices in the REI, as shown in Eqs. (1)–(4), was based on the relevance of road infrastructure, demographics, and hazard dimensions to exposure-based studies, as shown by El Rashidy and Grant-Muller [50]. Birkmann [21] emphasises that meaningful risk assessment must link hazard intensity with the exposure of populations and the condition of physical systems. His work supports the use of composite indicators when detailed network data are limited, particularly in regions where risks are driven by social and environmental pressures. Similarly, Contreras et al. [3] adopt a multidimensional approach that combines infrastructure, social exposure, and hazard intensity indicators in exposure modelling. In addition, Quinn [36] argues that road density serves as a proxy for accessibility and mobility, which are critical for economic activity and social connectivity [1].

In this study, road density is not interpreted as an indicator of redundancy but as a proxy for infrastructure exposure in conflict zones. Unlike classical transport engineering where higher road density is associated with redundancy and reduced vulnerability, in conflict prone regions, road density functions differently. It represents the extent of infrastructure footprint exposed to hazards. As Contreras et al. [3], Birkmann [21], and Cutter et al. [22] argue,

exposure in fragile environments must be operationalized as spatial intersection with disruptive hazards. In many Nigerian rural corridors, especially those affected by land-use competition, a denser road network does not necessarily indicate alternative routing capacity or resilience. Instead, it may indicate a larger amount of infrastructure potentially exposed to disruption. Studies by the Organisation for Economic Co-operation and Development/Sahel and West Africa Club (OECD/SWAC) [35] on conflict-prone regions noted that the presence and visibility of infrastructure can elevate exposure to disruption because roads act as strategic flashpoints, mobility channels for pastoral groups, and focal points for competition [1, 36]. For this reason, road density is treated here as an exposure indicator consistent with approaches applied in fragile environments rather than as a surrogate for redundancy, which classical transport engineering assumes in well-connected urban networks.

Population density, on the other hand, significantly determines exposure levels in risk assessments, as densely populated areas are more likely to suffer significant human and economic losses during conflict-induced disruptions [33]. In the context of HFCs, population density also correlates with land pressure and resource competition, which are known drivers of conflict escalation [51]. Conflict density is included as a direct indicator of hazard intensity and recurrence within each region. Krätli and Toulmin [52] support the use of conflict-related variables as core inputs in spatial risk models, while El Rashidy and Grant-Muller [50] emphasize their value in identifying hotspots and anticipating future risk patterns. This approach aligns with contemporary risk-modelling practices that emphasize the interplay between hazard, exposure, and infrastructure resilience [9, 53]. Overall, the inclusion of road density, population density, and conflict density enables the REI to identify locations and road types that are more exposed to HFCs within the selected NPCs, while also reflecting the contextual fragility of these corridors.

2.1.1 Population density

Population density was calculated as the number of inhabitants per km² [54], as given by Eq. (1):

$$Pd = \frac{P}{A} \quad (1)$$

where, Pd denotes population density, P represents the total population, and A represents the land area used consistently in Eqs. (1)–(3). As pointed out by Jha et al. [55], population density is critical in understanding vulnerability to adverse conditions, such as floods.

2.1.2 Conflict density

Conflict density was calculated by adapting the population density formula to represent the number of HFC events per unit area, as shown in Eq. (2):

$$Cd = \frac{C}{A} \quad (2)$$

where, the conflict density is given as Cd , and C represents the total number of HFC events recorded within the geographical area.

2.1.3 Road density

Similarly, the road density was defined as the total length of roads per km² of land area using Eq. (3):

$$Rd = \frac{RL}{A} \quad (3)$$

where, Rd denotes road density, and RL is the road length.

2.2 The Road Conflict Exposure Index Model

The REI integrates Rd , Pd , and Cd into a single composite indicator representing socio-spatial exposure within each corridor [33, 50, 51]. Because the three indicators are measured on different scales and may be correlated, the variables were standardized and principal component analysis (PCA) was used to examine the relative contribution of each indicator to the overall variance structure [56]. PCA is appropriate for index construction where indicators overlap conceptually and statistically, because it transforms correlated variables into a smaller set of uncorrelated components that capture maximum variance in the data [56, 57].

First, for each NPC and for each conflict year considered, Rd , Pd , and Cd were computed at the ward level. To ensure comparability, each indicator was standardized using z -scores [57], as shown in Eq. (4):

$$Z_{ij} = \frac{X_{ij} - \mu_j}{\sigma_j} \quad (4)$$

where, X_{ij} is the value of indicator j (Rd , Pd , Cd) in ward i , and μ_j and σ_j are the mean and standard deviation of indicator j across wards (within the corridor dataset).

PCA was then conducted to decompose the correlation matrix R of the standardized variables, as given in Eq. (5):

$$Ra_k = \lambda_k a_k \quad (5)$$

where, $k = 1, 2, 3$ indexes the principal components, λ_k is the eigenvalue representing the amount of variance explained by component k , and a_k is the corresponding eigenvector. Each principal component is a linear combination of the original indicators (Rd , Pd , Cd), and the loadings ℓ_{jk} represent the contribution of indicator j to component k , as shown in Eq. (6).

$$\ell_{jk} = a_{jk} \sqrt{\lambda_k} \quad (6)$$

Because the first principal component (PC1) explained only 42.36% of the total variance, while PC2 and PC3 explained 33.42% and 24.22%, respectively, no single component dominated the variance structure. Therefore, relying only on PC1 would have biased the index toward a single latent dimension [58]. To avoid this, a variance-weighted squared-loading approach was first used as a diagnostic step to assess the contribution of each indicator across all components [59].

Specifically, each variable's contribution was computed as the sum of its squared loadings across all components, weighted by each component's proportion of explained variance. This approach captures the total variance contribution of each indicator while ensuring non-negative weights and preserving interpretability [59]. Thus, the weight assigned to variable j was computed using Eqs. (7)–(10):

$$w_j = \sum_{k=1}^m \pi_k \ell_{jk}^2 \quad (7)$$

where, w_j is the weight assigned to variable j , and π_k is the proportion of total variance explained by component k , defined as

$$\pi_k = \frac{\lambda_k}{\sum_{k=1}^m \lambda_k} \quad (8)$$

where, m is the total number of components. Because the three indicators were standardized, the total variance was equal to 3:

$$\sum_{k=1}^3 \lambda_k = 3 \quad (9)$$

Thus,

$$\pi_k = \frac{\lambda_k}{3} \quad (10)$$

The squared-loading contribution analysis revealed near-equal contributions of approximately 0.333 for Z_{RD} , Z_{PD} , and Z_{CD} . This result indicates that road density, population density, and conflict density contribute almost equally to the overall variance structure. Since the PC1 results did not provide sufficient empirical justification for assigning differential weights, an equal-weighted REI was adopted for the final index construction [57, 60].

Hence, the normalized weights are expressed as:

$$\tilde{w}_j = \frac{w_j}{\sum_{j=1}^p w_j}; \tilde{w}_{RD} \approx \tilde{w}_{PD} \approx \tilde{w}_{CD} \approx 0.333 \quad (11)$$

The REI was therefore calculated in Eq. (12):

$$REI_i = \sum_{j=1}^3 \tilde{w}_j Z_{ij} \quad (12)$$

Given the near-equal PCA contributions, the final equal-weighted REI is expressed as:

$$REI_i = 0.333 \times Z_{RD} + 0.333 \times Z_{PD} + 0.333 \times Z_{CD} \quad (13)$$

Table 2 presents the PCA diagnostic metrics used to support the adoption of equal weights in the final REI construction.

Table 2. Principal component analysis (PCA) model metrics

Variable / Statistic	Component 1	Component 2	Component 3	Variance-Weighted Contribution
Z_{RD}	0.7046	0.1080	−0.7013	≈0.333
Z_{PD}	0.7081	−0.0432	0.7048	≈0.333
Z_{CD}	−0.0458	0.9932	0.1069	≈0.333
Eigenvalue	1.2709	1.0026	0.7265	
Variance proportion	0.4236	0.3342	0.2422	
Cumulative proportion	0.4236	0.7578	1.0000	

Note: Total trace = 3.00; $N = 741$ observations; Z_{RD} = standardized road density; Z_{PD} = standardized population density; Z_{CD} = standardized conflict density.

A higher REI value indicates that a ward lies in a zone where the infrastructure footprint, population exposure, and conflict intensity jointly concentrate more strongly. This index should not be interpreted as a probability of conflict occurrence or as a direct measure of structural road vulnerability. Rather, it quantifies relative socio-spatial exposure intensity across wards within each corridor and conflict year [39].

After computing REI_i , values were optionally transformed to a 0–100 scale for interpretability using min–max normalization, as shown in Eq. (14):

$$REI_i^* = \frac{REI_i - \min(REI)}{\max(REI) - \min(REI)} \times 100 \quad (14)$$

The resulting boxplot summarizes the normalized REI values and supports the interpretation of relative exposure levels across wards and corridors.

2.2.1 The kernel density model

The REI values served as inputs to the GIS platform, and ArcGIS Pro’s Kernel Density tool was used to generate continuous spatial intensity surfaces of road exposure across the study corridors [61]. Specifically, the kernel density model was applied to identify the spatial concentration of normalized REI values within each corridor. Eq. (15) describes the kernel density model:

$$\text{Kernel density } (s) = \frac{1}{r^2} \sum_{i=1}^n \left[\frac{3}{\pi} \cdot \text{pop}_i \left(1 - \left(\frac{d_i}{r} \right)^2 \right)^2 \right] \quad (15)$$

where, $i = 1, \dots, n$ are the input points, pop_i is the population field value of point i , d_i is the distance between the points i and the location s where density is estimated, and r is the search radius. In the ArcGIS Kernel Density tool, the “population field” was therefore assigned the normalized REI value which served as the weight for each represented event to reveal the spatial concentration or sparsity of the REI. The estimated density was calculated for each raster cell within the specified corridor.

Kernel density was applied to the ward-level normalised REI values to generate continuous spatial intensity surfaces of road exposure along each conflict corridor [62, 63]. All analyses were conducted using ArcGIS Pro’s Kernel Density tool (version 3.0.2) [64] with the default quadratic Epanechnikov kernel under a projected coordinate system to support Euclidean distance calculations in meters and reduce distortion associated with geographic coordinates. Although Nigeria spans multiple UTM zones, the selected corridors are concentrated within zone directly attached to central Nigeria where distortion under Zone 32N is minimal at the study scale. A planar method was adopted following projection.

Following Thompson et al. [65], who used average nearest neighbour (ANN)-derived observed mean distance (OMD) as an empirical anchor for KDE search radius selection, bandwidths in this study were defined proportionally to corridor-specific OMD values presented in Table 3.

To empirically ground this selection, ANN analysis was conducted for each corridor to determine the mean inter-centroid spacing of ward locations [65, 66]. ANN returns the OMD, representing the average Euclidean distance from each ward centroid to its nearest neighbouring centroid, and the nearest neighbour index (NNI), which indicates whether the point pattern is clustered ($NNI < 1$), random ($NNI \approx 1$), or dispersed ($NNI > 1$). All corridors exhibited statistically significant spatial dispersion ($NNI > 1$, $p < 0.001$), confirming that ward centroids are not randomly distributed. Observed mean nearest neighbor distances ranged from 5.27 km in the Kaduna–Plateau (NW–NC)

corridor and 6.04 km in the Benue–Enugu (NC–SE) corridor to 10.83 km in the Nasarawa–Benue (NC–NC) corridor and 10.70 km in the Taraba–Benue (NE–NC) corridor.

An output cell size of 500 m was applied uniformly across all corridors. This cell size was selected because it provides sufficient spatial detail for corridor-level visualization while remaining consistent with the ward-level scale of the REI data, following the sensitivity analysis. Empirical studies suggest that cell size mainly affects the visual resolution of kernel density estimation (KDE) output surfaces, whereas bandwidth selection has a stronger influence on smoothing behavior and hotspot structure [67, 68]. The 500 m resolution was therefore considered appropriate for representing the smooth density surfaces generated by bandwidths of 8–13 km without introducing excessive raster detail that would be inconsistent with the administrative scale of the exposure index.

Table 3. Average nearest neighbour analysis and selected kernel density estimation (KDE) bandwidths

Corridor	Observed Mean Distance (km)	Nearest Neighbour Index (NNI)	z-Score	p-Value	Candidate Bandwidth (km)	Bandwidth / Mean Spacing	Selected Primary Bandwidth (km)
Kaduna–Plateau (NW–NC)	5.27	1.437	4.349	0.000	6	1.14×	8
					8	1.52×	
					10	1.90×	
Benue–Enugu (NC–SE)	6.04	1.454	4.170	0.000	6	0.99×	8
					8	1.32×	
					10	1.66×	
Nasarawa–Benue (NC–NC)	10.83	1.496	4.240	0.000	11	1.02×	13
					13	1.20×	
					15	1.39×	
Taraba–Benue (NE–NC)	10.70	1.346	3.806	0.000	11	1.03×	13
					13	1.21×	
					15	1.42×	

Note: NW = North West; NE = North East; SE = South East; NC = North Central.

Spatial bandwidth (search radius) is the most critical parameter in kernel-based methods, as it governs the trade-off between local detail and regional smoothing [69] and represents a critical modelling decision [64, 66, 69]. The literature in Table 4 reveals no universal bandwidth–cell size rule and reported ratios vary widely depending on study scale and objective. Notably, some studies used OMD, some relied on nearest neighbour (NN) distance multiples, some used regression fit, and others selected bandwidths based on visual clarity or scale-based reasoning. Consequently, bandwidth selection remains subjective and context dependent [66], suggesting that experimentation is recommended by Bailey and Gatrell [66] and OMD can be used as an empirical anchor [65].

This approach aligns with the principle that bandwidth should reflect the characteristic spatial scale of the underlying process, neither so small as to produce fragmented, discontinuous surfaces nor so large as to obscure meaningful spatial variation [64, 66]. Thus, bandwidths ranging from approximately 1.0 to 1.5 times the observed mean spacing were evaluated with a uniform output cell size of 500 m. The selected bandwidths were based on sensitivity testing that extended beyond the 1:1 specification used by Thompson et al. [65]. In this study, bandwidth-to-cell-size ratios ranged from 16:1 to 26:1, which falls within the range reported in previous KDE applications summarized in Table 4.

Accordingly, three spatial bandwidths were examined per corridor for sensitivity testing [69], presented in Table 5. Sensitivity testing across lower, medium, and higher bandwidths confirmed expected KDE behaviour, notably, smaller bandwidths produced sharper, more fragmented hotspot surfaces, while larger bandwidths yielded broader regional clusters with reduced local contrast [65, 66]. The selected primary bandwidths provided the most interpretable corridor-aligned exposure surfaces while maintaining hotspot stability across years and corridors.

The REI values combine HFC hazard occurrence, road infrastructure density, and population density around socio-spatial stress points. These values were treated as spatially distributed exposure intensities and interpolated across the NPCs using KDE to visualize corridor-level exposure patterns [35, 39]. The workflow of the REI calculation and KDE-based spatial visualization procedure is shown in Figure 3.

Table 4. Literature-based justification for bandwidth–cell size ratios

Author	Grid / Cell Size (m)	Bandwidth (m)	Ratio (Bandwidth: Cell Size)	Selection Basis
Thompson et al. [65]	1	Observed mean distance (OMD)	OMD: 1 m	Bandwidth set equal to OMD from average nearest neighbour (ANN).
Xie and Yan [70]	5, 10, 50, and 100	20, 100, 250, 500, 1000, and 2000	4:1 to 20:1	Visual interpretation of smoothing effect.
Okabe et al. [71]	Not reported	50, 100, 150, 200, 210, 220, and 230	Not applicable (N/A)	Based on study objective.
Anderson [72]	100	200	2:1	Bandwidth = $2 \times$ grid size; acknowledges subjectivity.
Erdogan et al. [73]	500	500	1:1	Used 0.5 km grid; quadratic kernel.
Erdogan et al. [74]	50	700	14:1	Reported in Srikanth and Srikanth [75] literature table.
Blazquez and Celis [76]	100	1000	10:1	Large bandwidth relative to grid size.
Thakali et al. [77]	400	400 and 800	1:1; 2:1	Tested equal and double bandwidth.
Hashimoto et al. [78]	Not reported	250	N/A	Selected bandwidth using highest adjusted R^2 after 50 m interval testing.
Ha and Thill [79]	Not reported	400; 800	N/A	Applied two bandwidths to capture fine and coarse structures.
Keskin et al. [80]	Not reported	200	N/A	Fixed 200 m bandwidth for campus-scale clustering analysis.
Steiniger and Hunter [81]–Bear A	200	1,240; 2,600	6.2:1; 13:1	Rule-of-thumb/reference (href) and data-driven (hmttd) bandwidths; ecological movement tracks.
Steiniger and Hunter [81]–Bear B	400	4,900; 3,250	12.25:1; 8.13:1	href and hmttd bandwidths; ecological movement tracks.
Brimicombe [82]	Not reported	$6 \times$, $9 \times$, $12 \times$ median nearest neighbour (NN) distance	Relative scaling only	Suggested multiples of median NN distance.
This study	500	8,000; 13,000	16:1; 26:1	OMD-anchored; sensitivity testing; constant cell size.

Table 5. Sensitivity analysis of kernel density estimation (KDE) parameters

Corridor	Scenario	Bandwidth (km)	Cell Size (m)	Observed Pattern Behaviour
Kaduna–Plateau (NW–NC) & Benue–Enugu (NC–NC)	Low (local scale)	6	500	Slight fragmentation; sharper ward transitions
	Medium (primary)	8	500	Continuous corridor-aligned hotspots; stable hierarchy
	High (regional)	10	500	Broader smoothing; reduced micro-variation
Nasarawa–Benue (NC–NC) & Taraba–Benue (NE–NC)	Low	11	500	Slightly fragmented surface
	Medium	13	500	Corridor-scale smoothing aligned with centroid spacing
	High	15	500	Mild oversmoothing but hotspot structure retained

Note: NW = North West; NE = North East; SE = South East; NC = North Central.

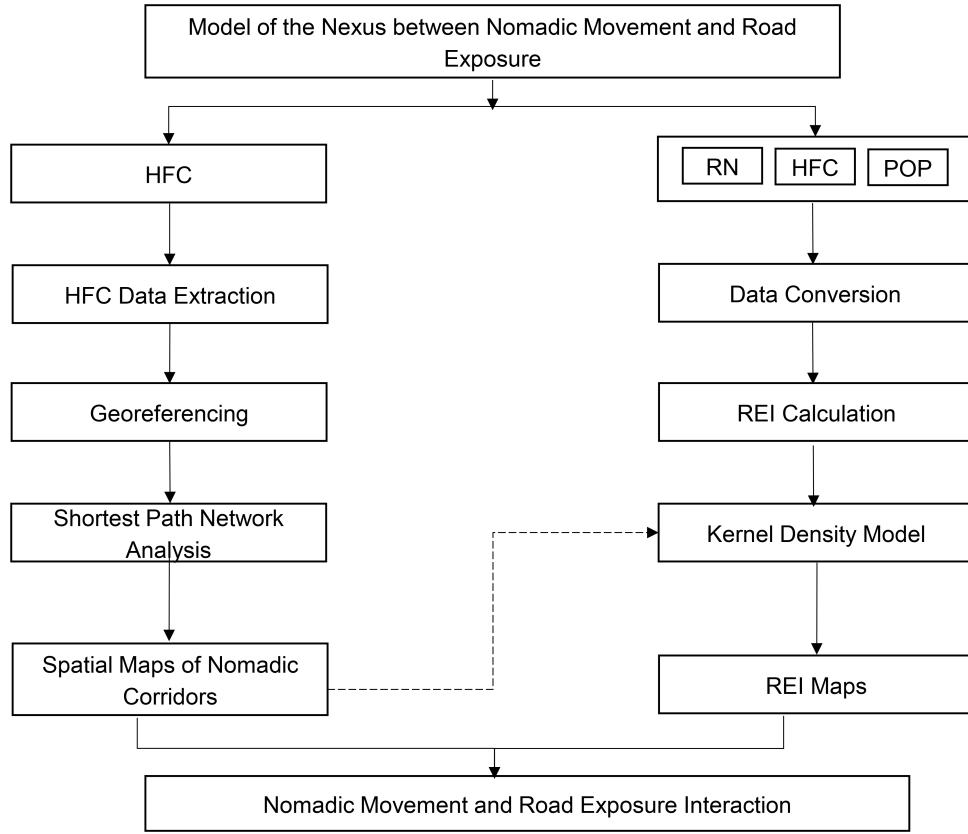


Figure 3. Workflow of the interaction between nomadic herder–farmer conflicts (HFCs) and road exposure in Nigeria

Note: RN = road networks; HFC = herder–farmers conflict; POP = population; REI = road exposure index.

2.3 Modifiable Areal Unit Problem Consideration

Because the exposure indicators were aggregated at the ward level, the results may be sensitive to the modifiable areal unit problem (MAUP), whereby statistical relationships can vary depending on the scale and zoning of spatial units [83, 84]. Wards were used because they are the smallest consistently available administrative units across all corridors and years, and they correspond to the scale at which local planning, security coordination, and development interventions are commonly implemented.

To maintain comparability, ward boundary definitions were kept consistent across the study period where possible. The KDE was subsequently applied to transform ward-level REI values into continuous spatial surfaces. This helps reduce abrupt visual boundary effects in choropleth mapping, although it does not eliminate the underlying MAUP. Accordingly, the REI should be interpreted as a relative exposure screening tool for inter-ward comparison rather than as a point-level risk estimate. Therefore, the results should not be used to infer individual-level exposure or precise site-specific risk.

2.4 Supplementary Analysis of Road-Class Exposure

To examine the association between road classes and the REI, a four-stage empirical strategy was implemented. These analysis are not intended as an independent validation of the REI, because road density is already one component of the index. Instead, it provides a supplementary assessment of which road classes are more strongly associated with higher REI values. The strategy included: (1) a panel regression of road-class densities on REI; (2) pairwise coefficient-equality tests to examine whether the effects of different road classes were statistically distinguishable; (3) corridor-level structural dominance analysis to assess the proportional contribution of each road class to total corridor road density; and (4) an analysis of variance (ANOVA) to determine whether wards dominated by different road classes exhibited different mean REI values.

Stage 1: Panel Regression of Road Density on REI

To assess whether road density by class significantly predicts exposure, a panel regression framework with year and corridor fixed effects was employed using ward–year observations (2015–2023). The panel specification allows exploitation of both cross-sectional variation (between wards) and temporal variation (within wards) while controlling for common time shocks and corridor-level structural heterogeneity. Thus, this specification improves the

interpretation of road-class associations with REI, although the results should be treated as supplementary because REI already incorporates overall road density.

The model is given in Eq. (16):

$$REI_{it} = \beta_1 PRD_{it} + \beta_2 SRD_{it} + \beta_3 TERD_{it} + \beta_4 TRD_{it} + \gamma_t + \delta_c + \varepsilon_{it} \quad (16)$$

where, REI_{it} = REI for ward i in year t , PRD_{it} = primary road density, SRD_{it} = secondary road density, $TERD_{it}$ = tertiary road density, TRD_{it} = trunk road density, γ_t = year fixed effects, δ_c = corridor fixed effects, ε_{it} = error term.

Stage 2: Pairwise Coefficient-Equality Tests

To test whether the associations between REI and different road classes were statistically distinguishable, Wald tests of coefficient equality were conducted within the same regression specification.

Formally, for two road classes r and s , the null hypothesis is shown in Eq. (17) with associated test statistics in Eq. (18):

$$H_0 : \beta_r = \beta_s \quad (17)$$

$$F = \frac{(\hat{\beta}_r - \hat{\beta}_s)^2}{\text{Var}(\hat{\beta}_r - \hat{\beta}_s)} \quad (18)$$

where, $\hat{\beta}_r$ and $\hat{\beta}_s$ are the estimated regression coefficients for two road classes r and s , $\text{Var}(\hat{\beta}_r - \hat{\beta}_s)$ is the variance of their difference.

These tests evaluate whether the estimated marginal effects (regression coefficients) of two road classes differ significantly. If the associated p -value exceeds 0.05, we fail to reject the null hypothesis of equality. In practical terms, this means that the observed difference between the two regression coefficients is not large relative to their statistical uncertainty and therefore cannot be considered statistically distinguishable at the 5% significance level.

Stage 3: Corridor-Level Structural Dominance Analysis

To examine whether spatial structure supports visual interpretation, the proportional share of each road class in total corridor road density was computed using Eq. (19):

$$\text{Share}_{rc} = \frac{\sum_i \text{Density}_{irc}}{\sum_i \sum_{q \in Q} \text{Density}_{iqc}} \quad (19)$$

where Share_{rc} is the proportional share of road class r in corridor c , Density_{irc} is the density of road class r in ward i within corridor c , q indexes each road class in the set of all road classes Q , and the denominator represents the total road density across all road classes and wards in corridor c .

Stage 4: ANOVA-Based Comparison of REI Across Dominant Road Classes

To test whether wards dominated by specific road classes exhibit higher realized exposure levels, wards were classified according to their dominant road class using Eq. (20):

$$\text{Dominant}_i = \arg \max_r (\text{Density}_{ir}) \quad (20)$$

where, Dominant_i denotes the dominant road class in ward i , and Density_{ir} is the density of road class r in ward i .

Mean REI values were compared across dominance groups using one-way ANOVA, as shown in Eq. (21):

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}} \quad (21)$$

where, MS_{between} denotes the between-group mean square and MS_{within} denotes the within-group mean square.

Post-hoc Bonferroni-adjusted pairwise comparisons were applied, as shown in Eq. (22):

$$p_{\text{adjusted}} = \min(p \times m, 1) \quad (22)$$

where, p is the original pairwise comparison p -value, m is the number of pairwise comparisons.

The homogeneity of variance was assessed using Bartlett's test, as shown in Eq. (23):

$$\chi^2 = \frac{(N - G) \ln(S_p^2) - \sum_{g=1}^G (n_g - 1) \ln(S_g^2)}{1 + \frac{1}{3(G-1)} \left(\sum_{g=1}^G \frac{1}{n_g - 1} - \frac{1}{N - G} \right)} \quad (23)$$

where, N is the total sample size, G is the number of groups, g is the group index, n_g is the sample size of group g , S_g^2 is the variance of group g , and S_p^2 is the pooled variance.

3 Results and Discussion

3.1 Maps of the Herder–Farmer Conflict Hotspot Corridors Using the Conflict Corridor Model

Following Faiyetole and Abiodun [10], a concentration of HFC events was identified in the NC geopolitical zone, which is bordered by all other geopolitical zones in Nigeria. This geographical position suggests that conflict dynamics in adjacent regions may influence, or be influenced by, those in the NC zone. These spatial patterns provide important contextual evidence for understanding the magnitude and distribution of HFC hotspots in the zone, as shown in Figure 4.

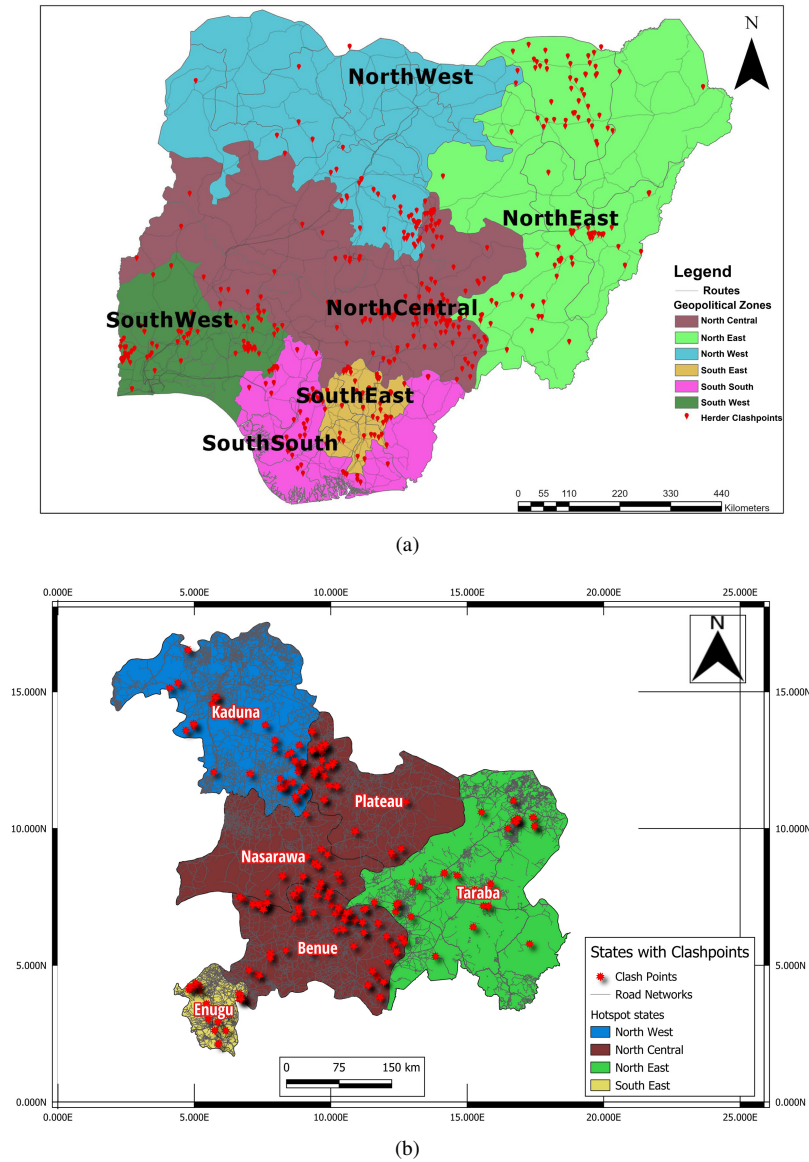


Figure 4. Geopolitical zones with herder–farmer conflict (HFC) hotspot states based on 2012 to 2023 HFC data: (a) the six geopolitical zones with HFCs; (b) six HFC hotspot states across four geopolitical zones, reproduced from Ref. [10] under the Creative Commons CC BY 4.0 license

To determine the possible NPCs, local government areas (LGAs) were selected based on their relative contributions to the total number of HFC events within each state. The shortest path method (SPM) was then applied to georeferenced HFC data and integrated with available road networks within the selected LGAs. This approach is based on the assumption that conflict locations may serve as spatial proxies for nomadic pastoral movement corridors, as illustrated in Figure 5, which depicts the shortest paths between pairs of clash points and is further elaborated in Figure 6. The four major NPCs identified include the Kaduna–Plateau (NW–NC), Benue–Enugu (NC–SE), Nasarawa–Benue (NC–NC), and Taraba–Benue (NE–NC) corridors [10]. Notably, these corridors are located in agrarian and major food-producing areas of the country, which is consistent with the findings of Madu and Nwankwo [85].

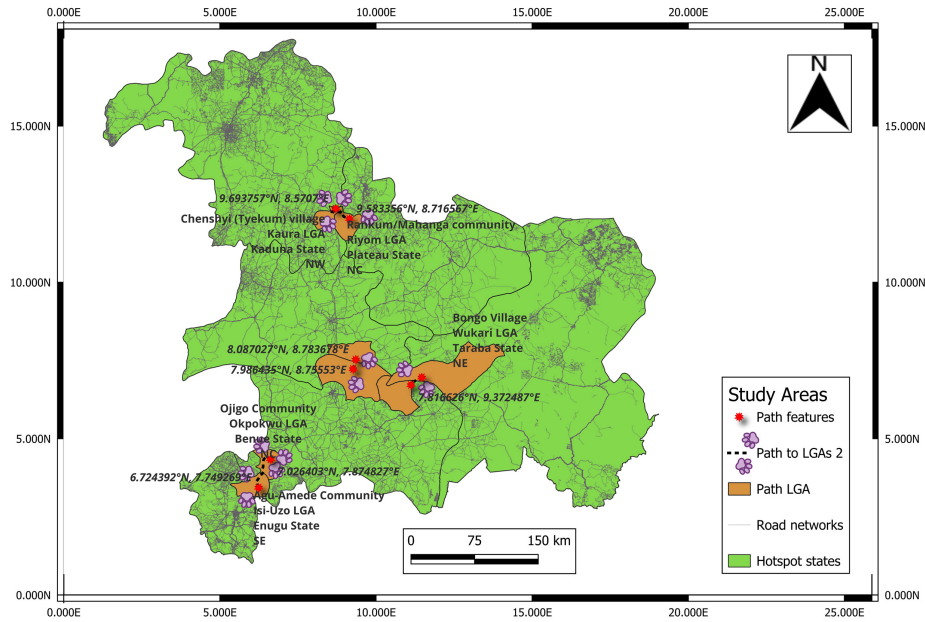


Figure 5. Herder–farmer conflict (HFC) hotspot corridors with local government areas (LGAs) and coordinates using the shortest path method (SPM), reproduced from Ref. [10] under the Creative Commons CC BY 4.0 license

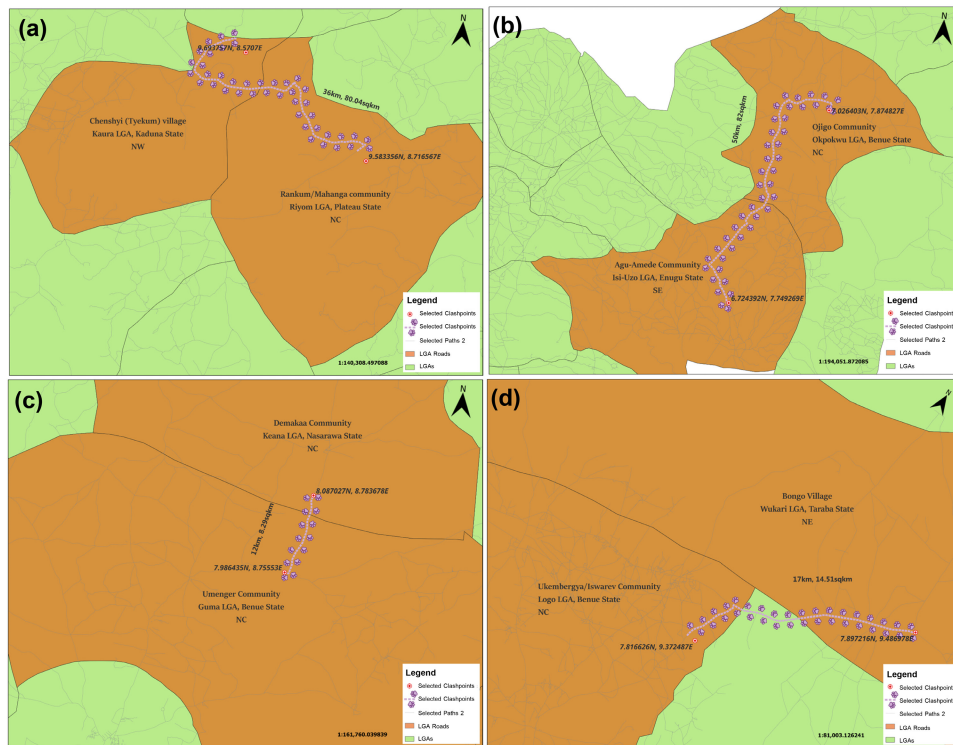


Figure 6. Four delineated herder–farmer conflict (HFC) hotspot corridors based on the shortest path method (SPM): (a) The Kaduna–Plateau (NW–NC) corridors; (b) Benue–Enugu (NC–SE) corridors; (c) Nasarawa–Benue (NC–NC) corridors; (d) Taraba–Benue (NE–NC) corridors, reproduced from Ref. [10] under the Creative Commons CC BY 4.0 license

Note: NW = North West; NE = North East; SE = South East; NC = North Central.

These results reveal similar spatial characteristics across the identified NPCs, as many of them are located in major food-producing areas [85–87]. This spatial overlap suggests that HFCs may have implications for food security, particularly through their potential effects on access to food-producing areas. According to the Food and Agriculture Organization (FAO) [88], food security consists of four dimensions: availability, access, utilization, and stability.

In a transportation context, HFCs in NPCs may disrupt access to rural food-producing areas by reducing road use, increasing perceived travel risk, or limiting the movement of farmers, traders, and transport operators. Such disruptions may, in turn, affect other dimensions of food security. For example, reduced access to rural production zones may constrain the movement of agricultural goods to urban markets, potentially affecting food availability and contributing to market instability. However, these links should be interpreted as potential pathways rather than direct causal effects, because additional empirical evidence would be needed to quantify the magnitude of such impacts.

The exposure of NPCs to HFCs may also be associated with shared agro-climatic and land-use conditions [11, 15, 89]. Previous studies have suggested that climatic stress, land-use change, and resource competition can contribute to conflict risk under certain social and institutional conditions [90–92]. In this context, examining the spatial relationship among HFC hotspots, pastoral movement corridors, road accessibility, and food-producing areas can help explain how localized conflict exposure may affect food-security risks at both local and broader scales. However, the mapped corridors should be interpreted as plausible conflict-exposure and mobility corridors rather than definitive routes of herder movement.

3.2 The Road Exposure Index Around the Herder–Farmer Conflict Hotspot Corridors

Following the OSM classification [93], four road types were identified in the study area: primary, secondary, tertiary, and trunk roads. Each corridor contains at least three of the four road categories. The REI was calculated at the ward level, and only years with actual conflict data were considered. For the Kaduna–Plateau (MW–NC) corridor, all wards within the Kaura and Riyom LGAs were considered and the conflict years started from 2017 to 2023. The Benue–Enugu (NC–SE) corridor examined all wards in the Okpokwu and Isi-Uzo LGA for 2017, 2018, 2020 and 2022. The Nasarawa–Benue (NC–NC) corridor treated all wards within the Keana and Guma LGA from 2017 to 2023 and the Taraba–Benue (NE–NC) corridor weighed all wards within the Logo, Ukum and Wukari LGA from 2015 to 2023.

The boxplot in Figure 7 indicates that the REI exhibits data sparsity across the NPCs for 2015 and 2016, reflecting the selection of conflict-occurrence years in the corridor-based analysis. For most years and corridors, the boxes and whiskers remain relatively low, mostly between 0 and 40%, indicating that REI values are low to moderate in most ward–year observations. However, 2018 shows the highest REI outliers, with values exceeding 75%, particularly in the Kaduna–Plateau (NW–NC) and Benue–Enugu (NC–SE) corridors. In contrast, Taraba–Benue (NE–NC) corridor consistently shows the lowest REI values, with median values generally around 10–15%, suggesting comparatively lower road exposure. The Kaduna–Plateau (NW–NC) and Nasarawa–Benue (NC–NC) corridors often exhibit slightly taller boxes and higher medians, around 15–25%, indicating higher or more variable road exposure than the other corridors. Overall, the corridors can be ranked from highest to lowest exposure as follows: Kaduna–Plateau (NW–NC), Nasarawa–Benue (NC–NC), Benue–Enugu (NC–SE), and Taraba–Benue (NE–NC).

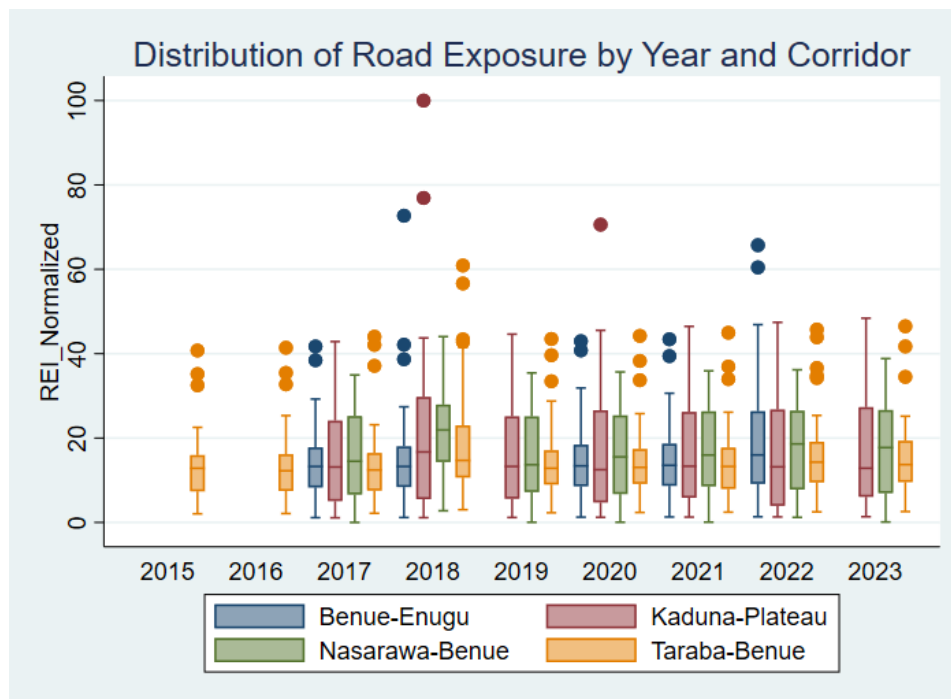


Figure 7. Normalized road exposure index (REI) values by year and corridor

3.2.1 Road exposure index for Kaduna–Plateau corridor between the North West and North Central geopolitical zones

The Kaduna–Plateau (NW–NC) corridor within Kaura and Riyom LGAs includes 27 wards presented in Figure 8 for the conflict years 2017 to 2023. The REI were categorized as follows: <10 (green band) indicates very low exposure; 10–20 (light green band) depicts low exposure; 20–30 (yellow band) represents medium exposure areas; 30–40 (orange band) shows high exposure; and >40 (red band) indicates critical or very high exposure around the NPCs. Wards in the medium to very high exposure categories indicate areas where roads and road users are more exposed to socio-spatial stress associated with HFCs.

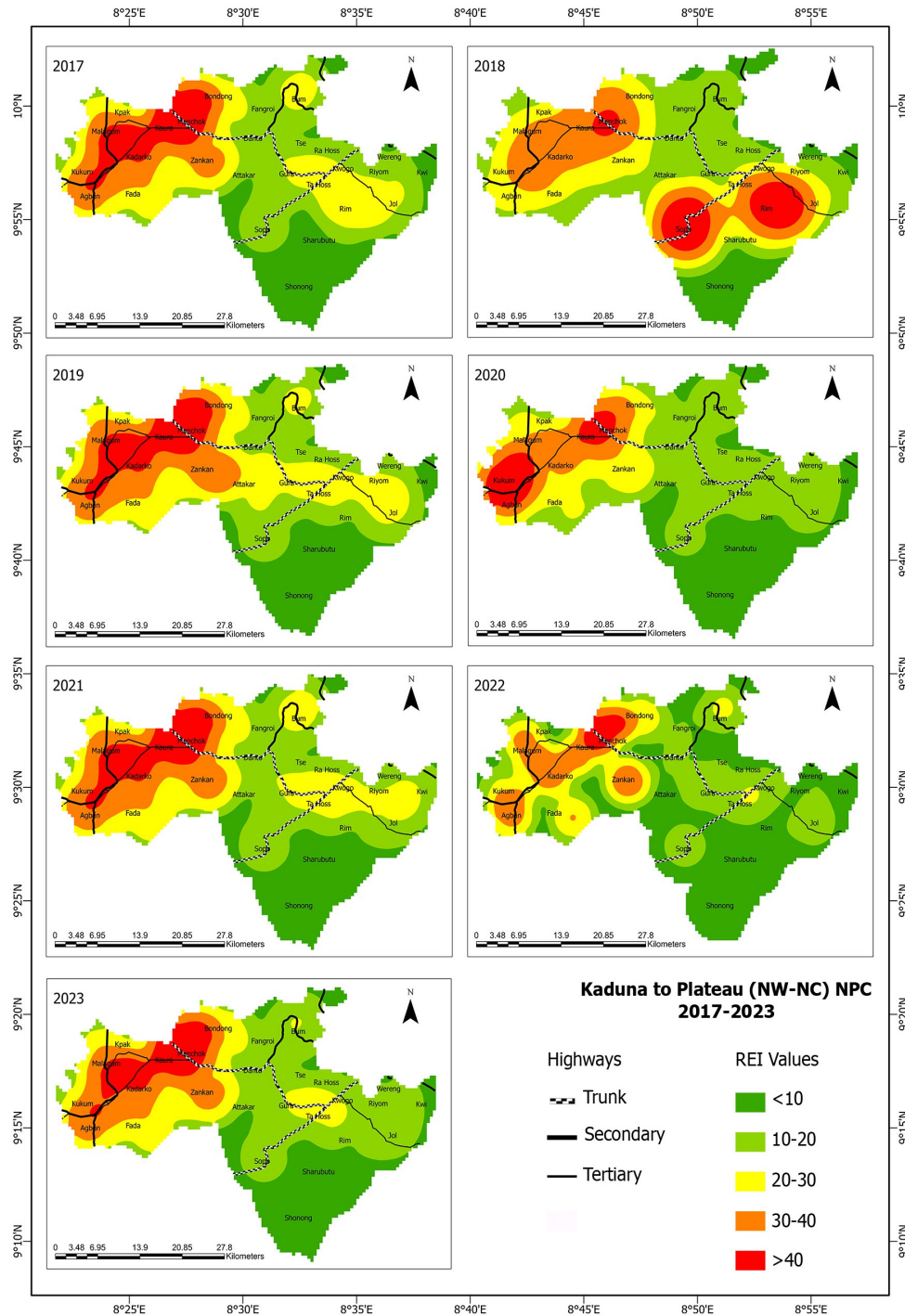


Figure 8. Road exposure index (REI) for the Kaduna–Plateau (NW–NC) corridor from 2017 to 2023

Note: NW = North West; NC = North Central.

The most notable feature of this corridor is the persistence of road exposure along the Kaduna, or northwestern,

axis, particularly in wards such as Agban, Fada, Kadarko, Kukum, Zankan, Manchok, Bondong, Kaura, Kpak, and Malagum. This pattern suggests that movement from the northwestern axis toward the northcentral axis is associated with sustained medium to high levels of road exposure.

In contrast, although the northwestern axis remains relatively stable in its high exposure pattern, a distinct localized spike appears in the Plateau axis in 2018. Specifically, Sopp and Rim wards in Riyom LGA exhibit isolated red-band values, indicating highly concentrated increases in REI values affecting road networks in the Plateau region during that year. The REI pattern also shows a gradual southward extension up to 2023, with most wards associated with medium to high exposure levels.

From 2017 onward, REI patterns increasingly align spatially with major transport corridors. Visual mapping shows that secondary, tertiary and trunk roads are frequently co-located with medium to very high REI zones. This pattern persists through 2023, when most wards in this NPC exhibit at least medium exposure levels, and the roads running from the northwestern to the northcentral axis are located within or near high-REI areas.

This spatial convergence reflects the mobility-driven nature of HFCs, which tend to concentrate along interconnected road networks rather than occurring randomly across space [35]. Given the hierarchical structure of the network (tertiary and secondary roads feed into trunk roads), disruptions along these axes can constrain inter-ward movement and amplify exposure across connected areas [94, 95]. As noted by UNDRR [96] and the IPCC [38], long-term exposure of critical infrastructure such as roads may weaken mobility, local coping capacity, and governance responses in affected areas.

3.2.2 Road Exposure Index for the Benue–Enugu corridor between the North Central and South East geopolitical zones

The Benue–Enugu (NC–SE) corridor includes 23 wards, as shown in Figure 9, for the conflict years 2017, 2018, 2020, 2021, and 2022. The REI pattern in this corridor shows a notable northward concentration, which gradually extends southward over time. The consistently most exposed areas in this corridor include Ugbokolo and Amejo wards, which show very high exposure from 2017 to 2022, while the remaining wards fluctuate between medium and high exposure levels. A general reduction in exposure is observed from the northcentral axis toward the southeastern axis. However, a gradual increase in REI values is also visible in the southwestern part of the corridor over time. The road types most frequently co-located with higher exposure are tertiary, secondary, and trunk roads. As the pattern progresses toward 2022, primary roads crossing the southwestern axis of this NPC also become more visually associated with higher REI values.

Consequently, the results reveal that, from 2017 to 2022, road junctions where multiple road types intersect are often located near high-exposure zones. This indicates that connectivity points may coincide with increased exposure, which is consistent with the argument that, in fragile settings, connectivity can magnify exposure to disruptive hazards [35].

3.2.3 Road Exposure Index for the Nasarawa–Benue corridor in the North Central geopolitical zone

The Nasarawa–Benue (NC–NC) corridor contains 20 wards presented in Figure 10 for the conflict years 2017 to 2023. A noticeable feature of this corridor is the concentration of exposure in the Keana LGA axis, although the corridor lies entirely within the North Central geopolitical zone. Movement southward is generally characterized by lower REI concentrations. Figure 10 shows that Galadima, Kardako, and Madaki wards are consistently associated with the highest REI during the study period, while many other wards remain within medium to high exposure categories. A progression southward is noticeable overtime. In contrast, Kaambe ward consistently records very low REI values despite being surrounded by medium- to high-exposure areas. This may be associated with the sparsity of roadways in the ward and the dominance of rice paddy farms along floodplains, whereas pastoral mobility is more likely to follow open pasture and upland areas [97].

In this NPC, tertiary, secondary and trunk roads are more associated with significant REI than primary roads, partly because primary roads are only marginally represented in this corridor.

3.2.4 Road Exposure Index for the Taraba–Benue corridor between the North East and North Central geopolitical zones

The Taraba–Benue (NE–NC) corridor contains 33 wards presented in Figure 11 for the conflict years 2015 to 2023. Unlike other corridors, movement southward in this corridor is associated with higher REI concentrations.

Exposure points are clustered mainly within the northcentral axis, which may suggest that the connection between the northeast and the northcentral zone increases exposure to HFC-related road stress. Figure 11 shows that Mbaker, Mbadyul, Tombo and Uyam wards consistently have the highest REI values from 2015 to 2023, with only a gradual and limited shift toward the northeastern axis of the corridor. Unlike other corridors, this NPC reveals a south-to-north movement pattern which may be attributed to the concentration of REI in the northcentral axis. Additionally, wards in the northeastern axis are more spatially dispersed, which may limit the spatial continuity of high-exposure clusters. The reported reduction in violent clashes in this area may also be associated with government-led and traditional interventions, although this interpretation requires further empirical support.

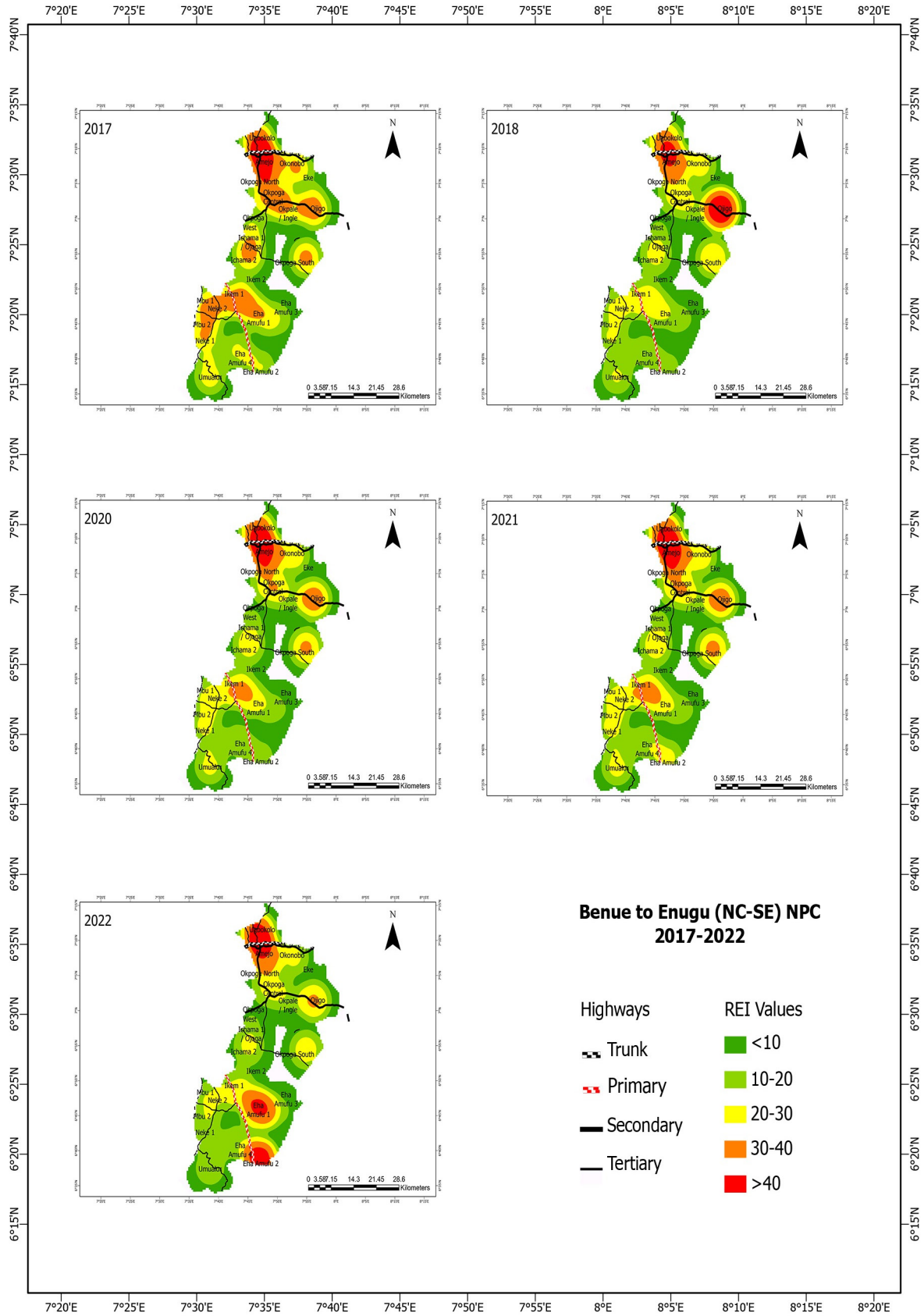


Figure 9. Road exposure index (REI) for the Benue–Enugu (NC–SE) corridor from 2017 to 2022

Note: NC = North Central; SE = South East.

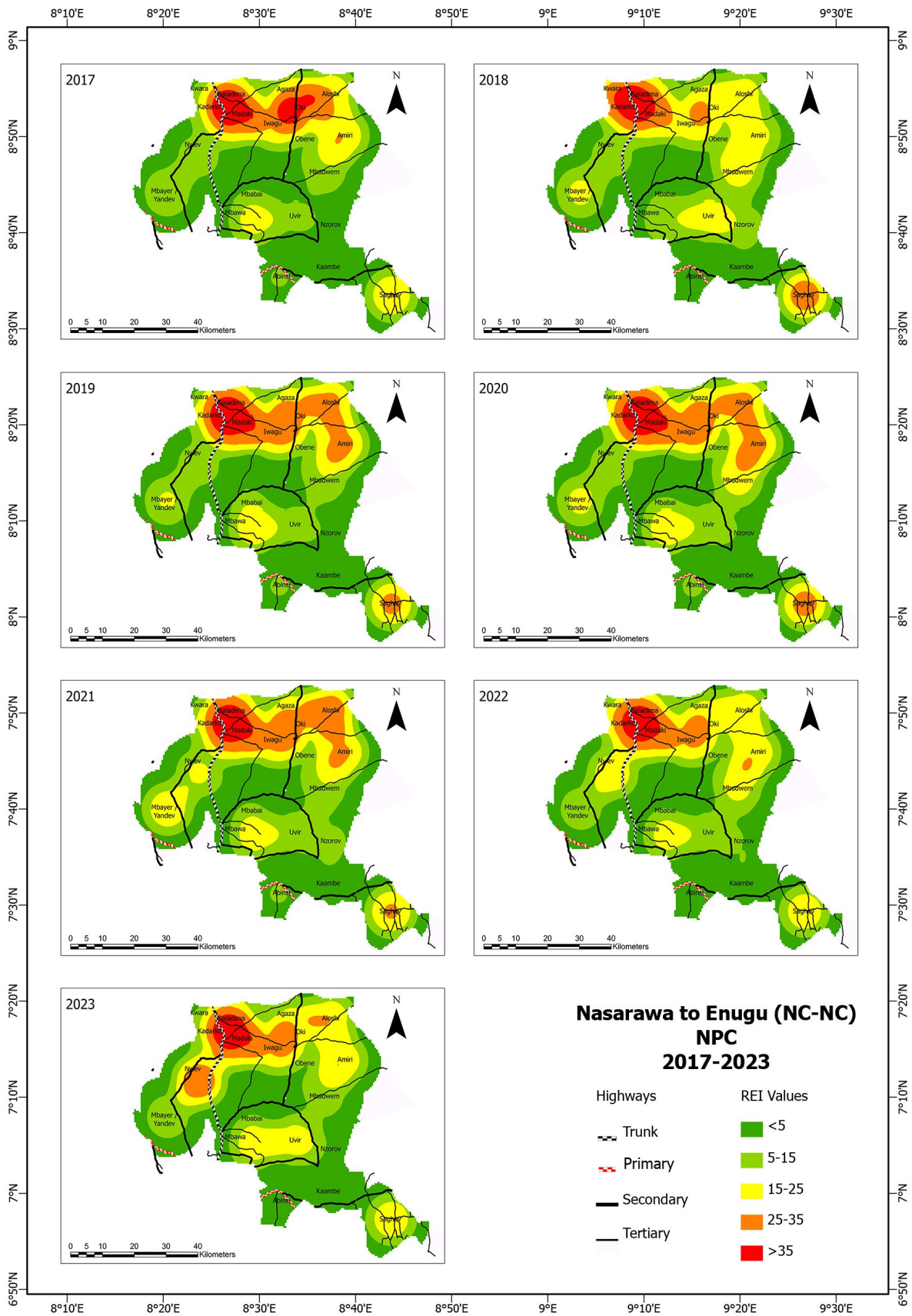


Figure 10. Road exposure index (REI) for the Nasarawa–Benue (NC–NC) corridor from 2017 to 2023

Note: NC = North Central.

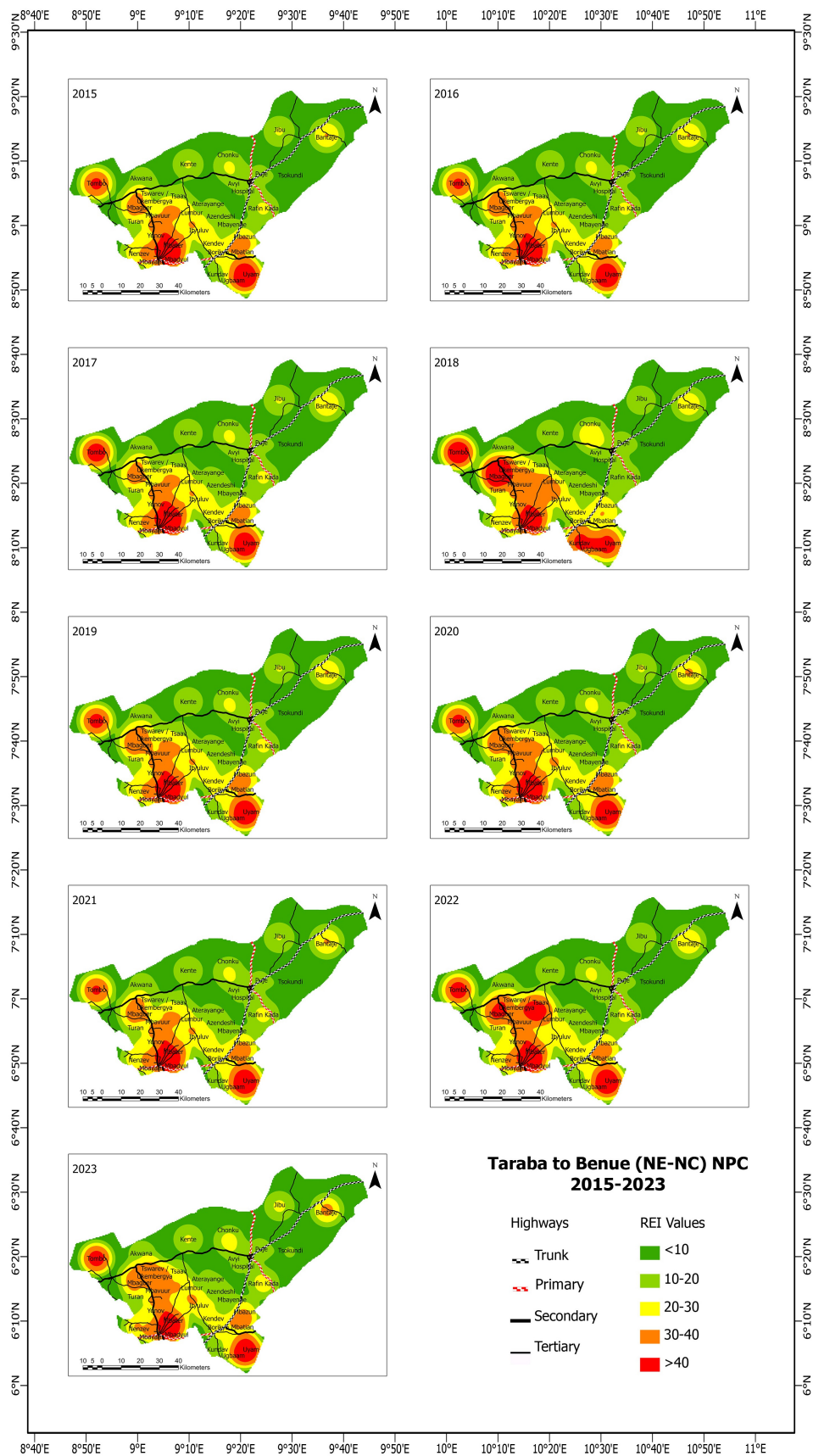


Figure 11. Road exposure index (REI) for the Taraba–Benue (NE–NC) corridor from 2015 to 2023

Note: NE = North East; NC = North Central.

In this NPC, all major highways are associated with medium to high exposure, however, the junction where tertiary, secondary and primary roads intersect are collocated with the highest REI values. Essentially, the visual inspection of the results further posits that tertiary, secondary and trunk roads are more exposed to HFCs in the four corridors than primary roads. In other words, primary highways appear to be associated with comparatively lower exposure in this dataset.

3.2.5 Road-class exposure patterns based on supplementary statistical analysis

To complement the visual interpretation of the REI maps, supplementary statistical results were used to examine how different road classes are associated with ward-level exposure. Table 6 shows that all road density classes are positively and significantly associated with REI (all $p < 0.001$) after controlling for year and corridor fixed effects. This indicates that higher road density is generally associated with higher exposure, regardless of road class.

Table 6. Regression coefficients for road density classes

Road Density Class	Coefficient	Std. Error	t-Score	p-Value	95% Confidence Interval
Trunk	1,234.56	206.06	5.99	<0.001	825.85–1,643.27
Primary	972.19	257.70	3.77	<0.001	461.05–1,483.33
Secondary	918.89	108.55	8.46	<0.001	703.57–1,134.21
Tertiary	793.78	47.35	16.76	<0.001	699.86–887.70

Note: Year and corridor fixed effects included; $N = 741$ observations; clusters = 103; coefficient of determination (R^2) = 0.543.

Pairwise tests coefficient tests in Table 7 show that the coefficients for primary, secondary, and tertiary roads are not statistically distinguishable from one another ($p > 0.26$). The only statistically significant difference is observed between trunk and tertiary roads ($p = 0.033$), while trunk roads are not significantly different from primary or secondary roads ($p > 0.19$). This indicates suggest that several road classes are positively associated with exposure, while differences among most classes are not statistically significant.

Table 7. Pairwise hypothesis tests for road-class coefficient equality

Comparison	F-Statistic	df	p-Value	Conclusion
Tertiary vs. Secondary	1.26	(1, 102)	0.264	Not significantly different
Tertiary vs. Primary	0.56	(1, 102)	0.456	Not significantly different
Secondary vs. Primary	0.04	(1, 102)	0.845	Not significantly different
Trunk vs. Secondary	1.72	(1, 102)	0.193	Not significantly different
Trunk vs. Primary	0.89	(1, 102)	0.347	Not significantly different
Trunk vs. Tertiary	4.67	(1, 102)	0.033	Significantly different
All classes jointly	92.81	(4, 102)	<0.001	All jointly significant

Note: df = degrees of freedom.

In addition, the corridor-level road-density shares in Table 8 provide additional context for interpreting these results. Tertiary roads constitute the largest share of total road density in three of the four corridors, accounting for approximately 47.9%–58.9% of total road density. Kaduna–Plateau (NW–NC) is the exception, where secondary, tertiary, and trunk roads have relatively similar shares. This pattern proposes that tertiary roads may contribute substantially to system-wide exposure because of their spatial prevalence within the corridor networks.

Table 9 further shows that wards dominated by tertiary roads have the highest mean REI value among the road-dominance groups. Bonferroni-adjusted comparisons indicate that tertiary-dominated wards differ significantly from other groups, whereas primary-, secondary-, and trunk-dominated wards are not significantly different from one another.

Consequently, these results indicate that tertiary-dominated wards exhibit significantly higher realized exposure levels. Tertiary roads therefore emerge as the most consistent contributor to elevated ward-level exposure, not because their marginal coefficient is uniquely larger, but because of their spatial ubiquity and dominance within corridor networks. This finding suggests that the relationship between road capacity and hazard exposure may depend on the type of hazard under consideration. In the context of HFC-related road exposure, elevated REI values are more pronounced around lower-capacity roads, especially tertiary roads, than around primary roads.

Table 8. Road-class density share and dominant class by corridor

Corridor	Trunk %	Primary %	Secondary %	Tertiary %	Dominant Class	Wards Dominant	Mean REI (Tertiary Wards)	Mean REI (All Wards)
Benue–Enugu (NC–SE)	10.8	13.1	28.4	47.7	Tertiary	10 of 23	17.30	17.35
Kaduna–Plateau (NW–NC)	33.4	0.0	33.7	32.9	Secondary	6 of 27	22.40	16.62
Nasarawa–Benue (NC–NC)	11.0	2.2	28.0	58.9	Tertiary	10 of 20	23.04	17.59
Taraba–Benue (NE–NC)	14.5	11.1	20.2	54.1	Tertiary	13 of 33	20.16	15.34

Note: Percentages may not sum to exactly 100.0 due to rounding; REI = road exposure index; NW = North West; NE = North East; SE = South East; NC = North Central.

Table 9. Comparison of ward-level road exposure index (REI) across dominant road classes

Dominant Road Class	Mean REI	Standard Deviation (SD)	95% Confidence Interval	N (Ward-Years)	Bonferroni Group	Exposure Rank
None / Zero	5.85	7.01	4.34–7.36	83	a	5
Trunk	16.84	14.13	14.23–19.45	113	b	2
Primary	15.20	8.87	13.09–17.31	68	b	3
Secondary	14.93	10.98	13.40–16.46	198	b	4
Tertiary	20.71	11.82	19.32–22.10	279	c	1
Overall	16.40	12.15	—	741	—	—

Note: Analysis of variance (ANOVA): $F(4, 736) = 29.21, p < 0.001$. Bartlett's test ($\chi^2(4) = 48.98, p < 0.001$) and robust variance tests (Levene/Brown–Forsythe; all $p < 0.001$) confirm that group differences persist despite heteroscedasticity. However, Bonferroni-adjusted comparisons show that tertiary-dominated wards differ significantly from all other dominant road classes. Groups sharing the same letter are not significantly different at $p < 0.05$.

3.2.6 Road exposure, border effects, and transport security in conflict-affected corridors

REI provides a framework for assessing the spatial exposure of conflict events, population, and road infrastructure. The results suggest that HFC-related road exposure is spatially uneven rather than randomly distributed, reflecting the interaction between pastoral mobility, settlement patterns, and road-network structure [9]. This approach is consistent with Esfeh et al. [6], who argued that road-disruptive hazards may affect not only specific road segments but also the surrounding areas and connected networks served by those roads. Hence, REI helps capture both road-network exposure and the surrounding exposure patterns within the NPCs. It should be noted that most corridors show a pattern of declining REI values southward, whereas the Taraba–Benue (NE–NC) corridor shows a different pattern, with exposure levels increasing toward the south. This differing pattern is consistent with studies suggesting that road presence can influence the spatial diffusion of hazards in different ways depending on context [1, 35]. Also, the results reveal a concentration of vulnerability in the northcentral axis of the country, except for the Kaduna–Plateau (NW–NC) corridor, where the northwestern axis records the highest REI values. The study further finds that tertiary roads, and in some cases trunk roads, play important roles in the NPC exposure structure, highlighting the importance of both feeder roads and higher-capacity roads in exposure analysis [6]. Even where higher-capacity roads are not directly exposed, their accessibility may be indirectly affected through disruptions to feeder roads and adjacent network links [26]. This suggests that the relationship between road capacity and hazard impact is context dependent and may vary by hazard type, road hierarchy, and local mobility patterns.

Notably, the concentration of high REI values along inter-state wards suggests the presence of a border effect across the delineated NPCs. Rather than interpreting this pattern merely as a spatial coincidence, it can be theoretically situated within frontier and borderlands scholarship, which conceptualizes administrative boundaries as socially and politically differentiated spaces where governance, surveillance, and resource regulation are often fragmented. Frontier zones are frequently characterized by overlapping jurisdictions, uneven institutional presence, and heightened contestation over land and mobility [92, 98]. In such liminal geographies, roads function simultaneously as mobility channels, economic lifelines, and strategic assets, thereby attracting both state control efforts and non-state contestation [35].

Nwankwo [99] observed that forested and border-adjacent zones along the Benue–Nasarawa–Taraba axis have evolved into spaces of limited formal control, facilitating both pastoral mobility and opportunistic violence. This aligns with broader evidence from the OECD/SWAC [35] report, which shows that violent events in West Africa disproportionately cluster near transport corridors that connect administrative regions, reflecting the strategic importance of roads in fragile border environments. Similarly, Jenelius et al. [8] argued that links connecting distinct territorial units often exhibit heightened exposure because their disruption generates spillover effects beyond the immediate locality. Starita and Scaparra [100] further showed that disruption at boundary links can propagate accessibility losses across adjacent jurisdictions, leading to demand shifts, route diversion, and suppressed mobility. In this sense, border corridors act as transmission nodes where localized conflict may translate into inter-regional transport insecurity. The REI captures this phenomenon empirically by identifying elevated exposure where road density, population pressure, and conflict intensity converge at geopolitical interfaces. Increased exposure in these corridors may disrupt transport flows between states and force commuters or freight operators to use less efficient alternative routes [100]. Additionally, such disruptions may contribute to shifts in mode choice, destination choice, and trip activity suppression [5, 100, 101].

Thus, measures to improve road security in exposed areas are necessary. Following Akanwa et al. [102], such measures may include the deployment of intelligent transport systems (ITS), improved road-network monitoring, and real-time communication with law-enforcement agencies along major exposed routes. These interventions may help reduce the effects of HFC-related road exposure and improve mobility resilience in conflict-affected corridors [103–105].

4 Conclusions and Recommendations

4.1 Conclusions

This study examined the spatial exposure of road infrastructure and road users to HFCs by developing a REI based on road density, population density, and conflict density. The REI provides a relative indicator of socio-spatial exposure across selected NPCs. The findings show that the selected corridors differ substantially in their REI values. From highest to lowest average exposure (see Table 8), the corridors are ranked as follows: Nasarawa–Benue (NC–NC) (17.59), Benue–Enugu (NC–SE) (17.35), Kaduna–Plateau (NW–NC) (16.62), and Taraba–Benue (NE–NC) (15.34).

Although the corridors vary in exposure intensity, high REI values are often concentrated around inter-state areas, suggesting a possible border effect in which road exposure may be shaped by overlapping administrative boundaries, mobility patterns, and conflict concentration. Such patterns may contribute to route diversion, trip suppression, and changes in travel behavior, although these outcomes require further empirical verification. The results further indicate that tertiary, secondary, and trunk roads are more frequently associated with elevated REI values than primary roads in the four corridors. This does not necessarily mean that these roads are directly disrupted or abandoned; rather, it suggests that lower-capacity feeder roads and some higher-capacity connecting roads may be more exposed to HFC-related socio-spatial stress within the NPCs. Overall, the study links nomadic pastoral mobility, road exposure, rural accessibility, and potential food-security risks in Nigeria, while emphasizing that the REI should be interpreted as a relative exposure-screening tool rather than a direct measure of vulnerability or road failure.

4.2 Recommendations: Targeted Intelligent Transport Systems Infrastructural and Security Interventions

The REI findings suggest the need for coordinated ITS and security interventions to reduce HFC-related transport risks across NPCs. In the short term, operational interventions could prioritize the rapid deployment of surveillance and monitoring technologies in high-exposure zones, particularly along inter-state borders and identified corridor hotspots. Mobile and solar-powered surveillance units, complemented by drone-based patrols, may provide real-time situational awareness in areas with limited infrastructure. These measures should be supported by basic early warning systems that disseminate timely information on emerging threats, unsafe routes, and alternative pathways to road users. Equally important is the establishment of community-based reporting networks, leveraging transport unions and local stakeholders as first responders to provide geotagged incident reports, thereby enhancing ground-level intelligence and response coordination.

Over the long term, a national ITS architecture could be developed to support corridor-based transport security and mobility resilience. This may include the development of fixed surveillance infrastructure supported by edge computing and artificial intelligence for anomaly detection, as well as the establishment of centralized traffic management centers that integrate multi-source data streams such as conflict records, mobility patterns, and environmental indicators. Predictive early warning systems could be developed to forecast conflict risks based on historical, climatic, and spatial data, enabling proactive rather than reactive interventions.

A critical strategic priority is the adoption of corridor-based security models that recognize NPCs as dynamic spaces of mobility, risk transmission, and economic significance. This entails classifying corridors based on risk levels, deploying targeted patrols, and, in the long run, establishing secure mobility corridors with dedicated infrastructure and coordinated inter-state governance mechanisms. Such coordination is particularly important in borderland

areas, where fragmented authority and overlapping jurisdictions may amplify HFC-related transport insecurity. By integrating surveillance technologies, early warning systems, community participation, and corridor-focused governance, these interventions may enhance transport resilience, improve accessibility, and mitigate the cascading effects of conflict exposure on food security and regional development.

5 Limitations and Future Study

The limited availability of officially recognized transhumance corridors nationwide, especially in southern Nigeria, informed the use of the SPM. Because SPM is deterministic, this represents a methodological limitation. However, the delineated NPCs are broadly consistent with existing literature and reports from international organizations, including Mercy Corps. The study also recognizes the possibility of reporting bias in the Nigeria Watch database and temporal instability in conflict patterns across some corridors. A further limitation relates to the reliance on OpenStreetMap-derived road layers, which may show uneven completeness across rural wards and may underrepresent tertiary roads or informal tracks. To reduce this limitation, major road classes were visually cross-checked using satellite imagery from Google Earth Pro. Additionally, the use of the KDE may overgeneralize exposure in areas where physical, administrative, or environmental barriers limit actual interaction between conflict events and road segments. This limitation is acknowledged as a trade-off for applying a consistent and interpretable method across large, data-scarce regions. Although empirically derived bandwidths were used to reduce subjectivity in kernel smoothing, the choice of output cell size remains partly subjective and may influence the visual representation of exposure surfaces.

Future studies could incorporate ground-truthing data, field surveys, traffic-flow data, and stakeholder interviews to validate the HFC time-series data and REI results. In addition, integrating agent-based modelling, mobility simulation, or dynamic spatial modelling could extend the CCM and provide a more detailed understanding of REI evolution over time.

Author Contributions

Conceptualization, A.A.F. and W.O.A.; methodology, A.A.F. and W.O.A.; software, A.A.F. and W.O.A.; validation, A.A.F. and W.O.A.; formal analysis, W.O.A.; investigation, A.A.F. and W.O.A.; resources, A.A.F.; data curation, A.A.F. and W.O.A.; writing—original draft preparation, W.O.A.; writing—review and editing, A.A.F. and W.O.A.; visualization, W.O.A.; supervision, A.A.F.; project administration, A.A.F. and S.O.B.; funding acquisition, A.A.F. All authors have read and agreed to the published version of the manuscript.

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Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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